Amrapali University, Haldwani

Faculty of Technology & Computer Applications

Department of Computer Science & Engineering

ChurnIQ — Customer Churn Prediction and Business Intelligence System

SUMMER INTERNSHIP PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the Award of

Bachelor of Technology (Computer Science & Engineering)

Academic Session: **2024-2028**

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Semester: 5
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Submitted To

Department of Computer Science & Engineering
Amrapali University, Haldwani

Internship Organization: **Aarsh Ai**

August 2026

CERTIFICATE

This is to certify that the Summer Internship Project Report entitled **“ChurnIQ — Customer Churn Prediction and Business Intelligence System”** is a bona fide record of the work carried out by **Ashish Dumka**, Enrollment Number **2410302008**, a student of **B.Tech CSE (AI&ML)**, Semester 5, Department of Computer Science & Engineering, Amrapali University, Haldwani.

The internship was carried out at **Aarsh Ai** in the role of Data Science from **11-07-2026** to **19-08-2026**, under the guidance of **Swasti Mishra**.

The report is submitted in partial fulfillment of the requirements for the award of the degree stated above. The work presented has not been submitted elsewhere for the award of any other degree.

Mrs. Sonali Mahajan	**Dr. Ashish Bhushan Khare**
Faculty Mentor	Head of Department

Date: 19-08-2026
Place: Haldwani

DECLARATION

I, **Ashish Dumka**, Enrollment Number **2410302008**, hereby declare that the Summer Internship Project Report entitled **“ChurnIQ — Customer Churn Prediction and Business Intelligence System”** submitted to Department of Computer Science & Engineering, Amrapali University, Haldwani, is an original record of the work carried out by me during my internship at Aarsh Ai.

The analysis, source code, results and conclusions presented in this report are my own work. Material drawn from other sources has been acknowledged in the references. This report has not been submitted for the award of any other degree or diploma.

Date: 19-08-2026
Place: Haldwani

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I express my sincere gratitude to my industry mentor, **Swasti Mishra** of **Aarsh Ai**, for the guidance and practical direction provided throughout the internship. The review sessions shaped both the technical decisions taken in this project and my understanding of how analytical work is applied in practice.

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ABSTRACT

Customer churn — a subscriber discontinuing a service — directly reduces recurring revenue, and retaining an existing customer is generally less costly than acquiring a new one. This report describes ChurnIQ, a customer churn prediction and business intelligence system developed during a Data Science internship at Aarsh Ai.

The system was built on the public Telco Customer Churn dataset of 7,043 customer records described by 21 attributes, in which 1,869 customers (26.54%) had churned. The data was validated, cleaned and passed through a preprocessing pipeline that imputes and scales numerical attributes and one-hot encodes categorical attributes. Six classification algorithms were trained and compared on identical partitions using accuracy, precision, recall, F1 score and ROC-AUC, supported by stratified five-fold cross-validation.

Random Forest achieved the highest F1 score (0.6226) and was persisted as the deployed model, while Gradient Boosting achieved the highest ROC-AUC (0.8437). Predicted probabilities are converted into a risk score, four risk levels and six strategic customer segments, from which a rule-based engine derives a recommended retention action and engagement channel for each customer. The results are presented through an eleven-page Streamlit dashboard supporting dataset upload, prediction, segmentation, customer search and CSV export.

The system identifies and prioritises customers who are statistically likely to leave; it does not guarantee that any customer will be retained. All figures reported here were produced by executing the repository code on the stated dataset.

Keywords: churn prediction, classification, scikit-learn, random forest, customer segmentation, risk scoring, business intelligence, Streamlit.

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CHAPTER 1
INTRODUCTION

1.1 Background

Subscription businesses such as telecommunications operators earn revenue on a recurring basis, so the departure of an existing customer removes a predictable future income stream. Acquiring a replacement customer generally costs more than retaining the existing one, which makes early identification of customers likely to leave commercially useful. Historical service records already contain the contract, billing and usage patterns associated with past departures, so the problem is well suited to supervised machine learning.

1.2 Problem Statement

Churn is usually detected only after the customer has cancelled, at which point intervention is no longer possible. A predictive model alone is also insufficient: a probability score does not tell a retention team which customers matter most or what action to take. The problem addressed here is therefore twofold — to predict churn reliably from historical data, and to convert those predictions into a prioritised, actionable output.

1.3 Objectives

Table 1.1: Objectives of the ChurnIQ project

No.	Objective
1	Validate and clean the raw customer dataset and report its quality.
2	Analyse the dataset to identify the attributes most associated with churn.
3	Build a reusable preprocessing pipeline for numerical and categorical attributes.
4	Train and compare several classification algorithms on identical partitions.
5	Select and persist the best model together with its preprocessing pipeline.
6	Convert predicted probabilities into risk scores, risk levels and customer segments.
7	Generate a recommended retention action and channel for each customer.
8	Present the results through an interactive dashboard with export facilities.

1.4 Scope

The system covers the complete analytical path from a raw CSV or Excel file to an exported retention report: validation, cleaning, exploratory analysis, preprocessing, model training and comparison, prediction, risk scoring, segmentation, recommendation and dashboard presentation. It operates on file-based data supplied by the user and does not perform live billing integration, automated campaign execution or user authentication.

1.5 Organisation of the Report

Chapter 2 describes the organisation and the internship. Chapter 3 covers the technologies used and Chapter 4 the system analysis. Chapter 5 presents the design and Chapter 6 the dataset and preprocessing. Chapters 7 and 8 cover exploratory analysis and model development. Chapters 9 to 11 describe segmentation, the dashboard and the business logic. Chapters 12 and 13 present testing and results, and Chapters 14 and 15 discuss limitations, future scope and conclusions.

CHAPTER 2
ORGANIZATION AND INTERNSHIP OVERVIEW

2.1 Internship Details

Table 2.1: Internship details

Field	Detail
Organization	Aarsh Ai
Role / Designation	Data Science
Industry mentor	Swasti Mishra
Start date	11-07-2026
End date	19-08-2026
Duration	Approximately 6 weeks
Department	Department of Computer Science & Engineering
Academic session	2024-2028

2.2 Nature of the Work

The internship was undertaken in the role of Data Science at Aarsh Ai. The work was project-based: a single analytical system was carried from problem definition through to a working dashboard, rather than isolated exercises. Progress was reviewed with the industry mentor, and technical decisions such as the choice of evaluation metric and the design of the segmentation rules were settled in those reviews.

2.3 Responsibilities

- Validating and cleaning the customer dataset and documenting its quality.
- Performing exploratory analysis to identify the strongest churn indicators.
- Building the preprocessing pipeline and training the classification models.
- Implementing the prediction, segmentation and recommendation modules.
- Developing the Streamlit dashboard and its export features.
- Testing the modules against valid, incomplete and malformed inputs.

2.4 Execution Plan

Table 2.2: Phase-wise execution plan

Phase	Activity	Deliverable
1	Problem study, dataset acquisition and project setup	Repository structure, raw dataset
2	Dataset validation and cleaning	dataset_validator.py, clean_data.py
3	Exploratory data analysis	eda.py
4	Feature engineering and preprocessing	feature_engineering.py, preprocessing.pkl
5	Model training, comparison and persistence	model_training.py, best_churn_model.pkl
6	Prediction pipeline and risk scoring	prediction_pipeline_backup.py
7	Segmentation and recommendations	segmentation.py, recommendations.py
8	Dashboard development, styling and testing	app.py, style.css

2.5 Learning Outcomes

The internship provided practical experience of an end-to-end machine learning workflow: handling imperfect real data, choosing evaluation metrics appropriate to an imbalanced problem, persisting a model together with its preprocessing so that predictions remain reproducible, and presenting analytical output in a form a non-technical user can act upon.

CHAPTER 3
TECHNOLOGIES AND TOOLS

3.1 Technology Stack

The system is written entirely in Python. The libraries below are those actually imported by the source files in the repository.

Table 3.1: Technology stack used in the project

Component	Technology	Role in ChurnIQ
Language	Python 3	All modules and the dashboard
Data handling	pandas, numpy	Loading, cleaning and aggregating tabular data
Machine learning	scikit-learn	Preprocessing pipeline, six classifiers, metrics, cross-validation
Visualisation	plotly, matplotlib, seaborn	Interactive dashboard charts and static analysis figures
Interface	Streamlit	Eleven-page web dashboard
Persistence	joblib	Saving and loading the model pipeline
File formats	openpyxl, xlrd	Reading .xlsx and .xls uploads
Database (optional)	mysql-connector-python	Optional MySQL storage in database.py
Styling	CSS	Custom dashboard theme in style.css

3.2 Rationale

pandas provides the dataframe structure used throughout, and numpy supports the numerical operations underneath it. scikit-learn was chosen because it supplies the estimators, the `Pipeline` and `ColumnTransformer` objects that bind preprocessing to the model, and the evaluation utilities in one consistent interface. Streamlit was chosen because it renders a Python script directly as a web application, which suited a project where the analytical code and the interface are maintained together.

3.3 Persistence and Reproducibility

joblib stores the fitted preprocessing pipeline and the trained estimator as a single object. Because the transformer is saved with the model, a new dataset passes through exactly the transformations used during training, which prevents the mismatch that occurs when preprocessing is reimplemented at prediction time.

CHAPTER 4
SYSTEM ANALYSIS

4.1 Existing Approach and Its Limitations

Table 4.1: Comparison of the existing approach with the proposed system

Limitation of manual analysis	How ChurnIQ addresses it
Churn is identified only after cancellation	Probability is estimated for every active customer
Spreadsheet review does not scale to thousands of rows	The full dataset is scored in a single pass
No consistent ranking of who to contact first	A numeric risk score and four risk levels order the customers
Revenue exposure is not quantified	Revenue at risk is computed per customer and per segment
Findings are not reusable	The model and pipeline are persisted and reloaded

4.2 Functional Requirements

Table 4.2: Functional requirements

ID	Requirement	Implemented in
FR-1	Upload a CSV or Excel dataset	app.py
FR-2	Validate structure and report data quality	dataset_validator.py
FR-3	Clean the dataset and handle missing values	clean_data.py
FR-4	Produce exploratory statistics and charts	eda.py
FR-5	Apply the preprocessing pipeline	feature_engineering.py
FR-6	Train and compare classification models	model_training.py
FR-7	Predict churn probability for every customer	prediction_pipeline_backup.py
FR-8	Assign risk scores, risk levels and segments	segmentation.py
FR-9	Generate retention actions and channels	recommendations.py
FR-10	Display results and export CSV reports	app.py

4.3 Non-Functional Requirements

- **Reliability:** a missing model file or malformed upload produces a message rather than an unhandled exception.
- **Usability:** the dashboard is operable without programming knowledge; every table can be exported as CSV.
- **Reproducibility:** a fixed random seed of 42 and a persisted pipeline make results repeatable.

- **Maintainability:** each stage is a separate module in `Src/` with a documented entry point.

4.4 System Requirements

Table 4.3: Hardware and software requirements

Type	Requirement
Processor	Dual-core 2.0 GHz or higher
Memory	4 GB minimum, 8 GB recommended
Storage	Approximately 500 MB including the saved model
Operating system	Windows, Linux or macOS
Software	Python 3, the packages listed in requirements.txt, a modern web browser

CHAPTER 5
SYSTEM DESIGN

5.1 Architecture

ChurnIQ is organised in layers. The presentation layer is the Streamlit application; the analytical layer contains the modules that clean, model, segment and recommend; the persistence layer holds the datasets and the saved model artefacts. The dashboard calls the analytical modules and never manipulates the model internals directly.

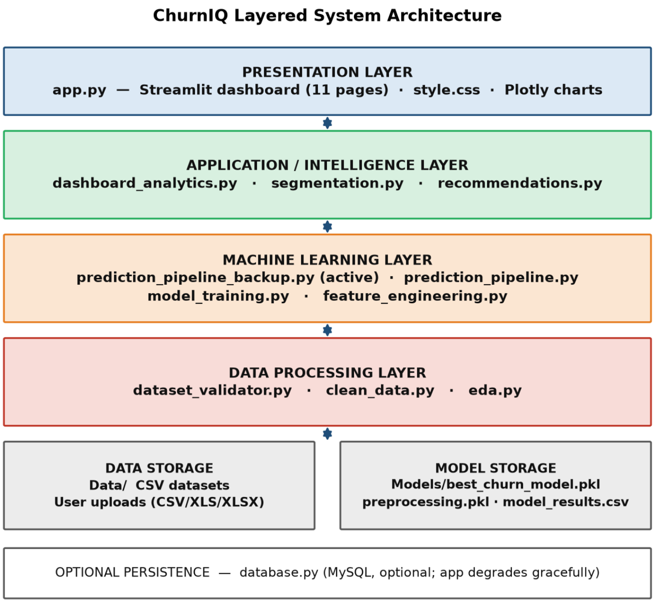

Figure 5.1: Layered system architecture

5.2 Workflow

A dataset enters the system by upload or by loading a default file, is validated and cleaned, then either scored by the saved model or, if the columns do not match, by one of two fallback strategies described in Chapter 8. The scored data is enriched with risk levels, segments and recommendations before being displayed.

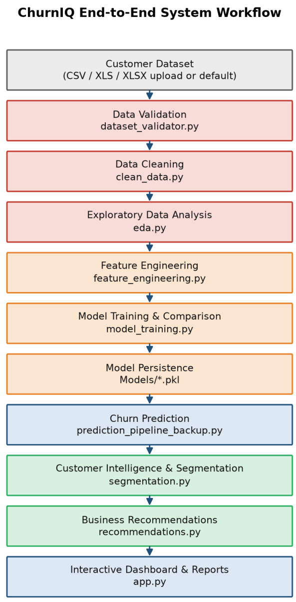

Figure 5.2: End-to-end system workflow

5.3 Module Structure

Table 5.1: Project modules and their responsibilities

Module	Lines	Responsibility
app.py	5,232	Streamlit dashboard, all eleven pages
prediction_pipeline_backup.py	2,783	Prediction, risk scoring, customer intelligence
dashboard_analytics.py	2,360	Aggregations and metrics used by the dashboard
feature_engineering.py	1,177	Preprocessing pipeline construction
segmentation.py	1,076	Risk levels and strategic segments
eda.py	981	Exploratory analysis routines
model_training.py	963	Training, comparison and selection
recommendations.py	959	Retention actions and channels
database.py	823	Optional MySQL persistence
dataset_validator.py	483	Structure and quality checks
clean_data.py	429	Cleaning and type correction

CHAPTER 6
DATASET AND DATA PREPROCESSING

6.1 Dataset Description

The reference dataset is the public Telco Customer Churn dataset, containing 7,043 customer records described by 21 attributes. Each row is one customer, identified by `customerID`, and the target attribute `Churn` records whether that customer left.

Table 6.1: Dataset summary computed from the repository data file

Property	Value
Records	7,043
Attributes	21
Target attribute	Churn (Yes / No)
Churned customers	1,869 (26.54%)
Retained customers	5,174 (73.46%)
Duplicate rows	0
Blank TotalCharges values	11

6.2 Attribute Groups

- **Demographic:** gender, SeniorCitizen, Partner, Dependents.
- **Account:** tenure, Contract, PaperlessBilling, PaymentMethod.
- **Services:** PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- **Financial:** MonthlyCharges, TotalCharges.

6.3 Data Quality and Cleaning

Validation reported no duplicate rows and no null values. Eleven records contain a blank string in `TotalCharges`; all belong to customers with a tenure of zero months, so no billing total exists yet. The cleaning module converts the column to a numeric type, which turns those blanks into missing values that the pipeline then imputes. The target is mapped from Yes and No to 1 and 0.

6.4 Preprocessing Pipeline

Preprocessing is defined as a scikit-learn `ColumnTransformer` so that every transformation is fitted on the training partition only and reapplied unchanged at prediction time.

Table 6.2: Preprocessing applied to each attribute type

Attribute type	Steps applied
Numerical (4 attributes)	Median imputation, then standard scaling
Categorical (15 attributes)	Most-frequent imputation, then one-hot encoding with handle_unknown='ignore'
Identifier	customerID excluded from the feature set

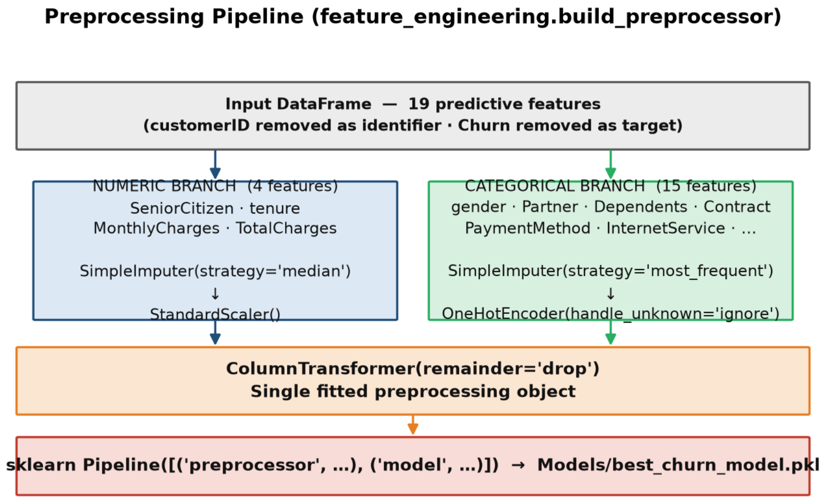

Figure 6.1: Preprocessing pipeline built by build_preprocessor()

6.5 Train and Test Partitions

The dataset is divided using a stratified split so that both partitions preserve the original churn proportion.

Table 6.3: Train and test partition parameters

Parameter	Value
Training records	5,634 (80%)
Testing records	1,409 (20%)
Stratified on	Churn
Random state	42
Features after encoding	19 input attributes

CHAPTER 7
EXPLORATORY DATA ANALYSIS

7.1 Churn Distribution

Of 7,043 customers, 1,869 (26.54%) churned. The classes are therefore imbalanced roughly three to one, which is the reason accuracy alone is not used to judge the models in Chapter 8.

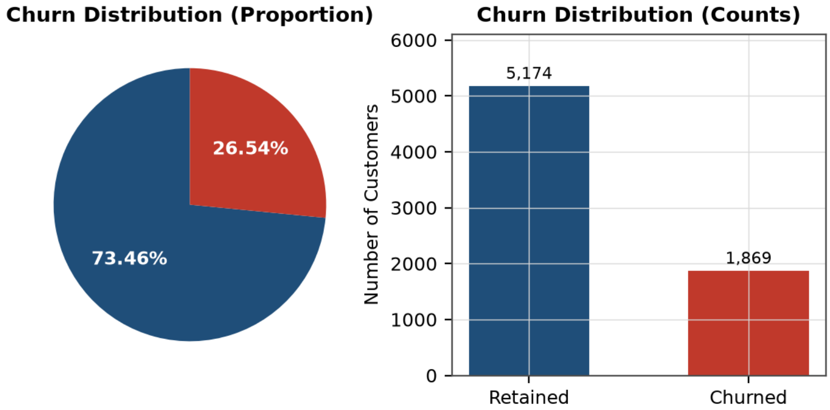

Figure 7.1: Churn distribution in the reference dataset

7.2 Contract Type

Contract length is the single strongest indicator in the dataset. Month-to-month customers churn at more than fifteen times the rate of two-year customers.

Table 7.1: Churn rate by contract type

Contract type	Customers	Churn rate
Month-to-month	3,875	42.71%
One year	1,473	11.27%
Two year	1,695	2.83%

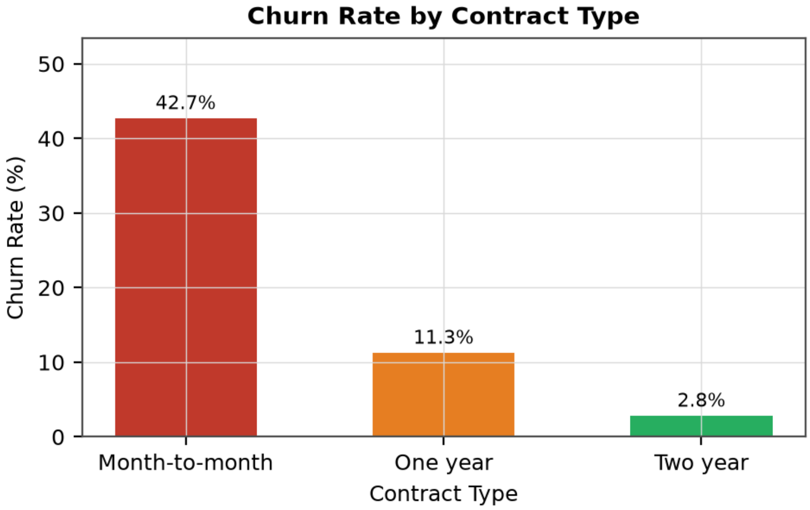

Figure 7.2: Churn rate by contract type

7.3 Payment Method

Customers paying by electronic check churn at 45.29%, far above the other three methods, which fall between 15% and 20%. The automatic methods show the lowest rates.

Table 7.2: Churn rate by payment method

Payment method	Customers	Churn rate
Electronic check	2,365	45.29%
Mailed check	1,612	19.11%
Bank transfer (automatic)	1,544	16.71%
Credit card (automatic)	1,522	15.24%

7.4 Tenure

Churn falls steadily as tenure increases: more than half of customers in their first six months leave, against 6.61% of those past sixty months. The first year is therefore the critical retention window.

Table 7.3: Churn rate by tenure group

Tenure group	Customers	Churn rate
0-6 months	1,481	52.94%
7-12 months	705	35.89%
13-24 months	1,024	28.71%
25-36 months	832	21.63%
37-60 months	1,594	16.62%
Over 60 months	1,407	6.61%

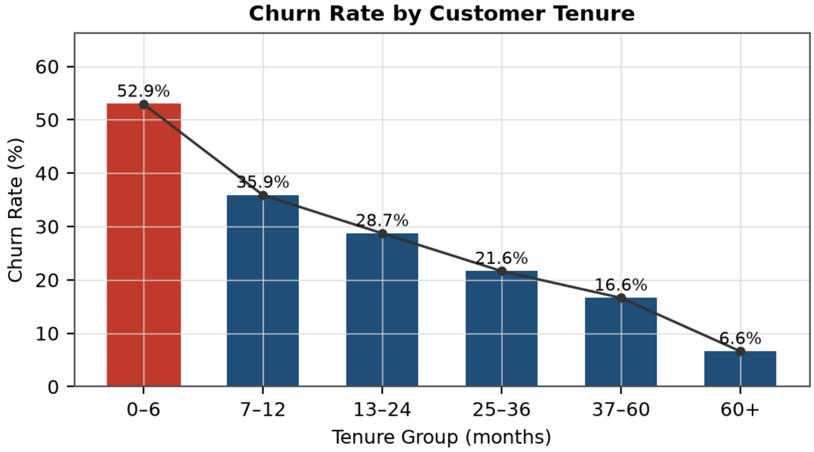

Figure 7.3: Churn rate by customer tenure group

7.5 Services and Demographics

Fibre optic subscribers churn at 41.89% against 18.96% for DSL. Customers without technical support churn at 41.64% against 15.17% for those with it, and the pattern repeats for online security (41.77% against 14.61%). Senior citizens churn at 41.68% against 23.61% for other customers, and customers without a partner or dependents churn more often. Gender shows almost no effect (26.92% against 26.16%).

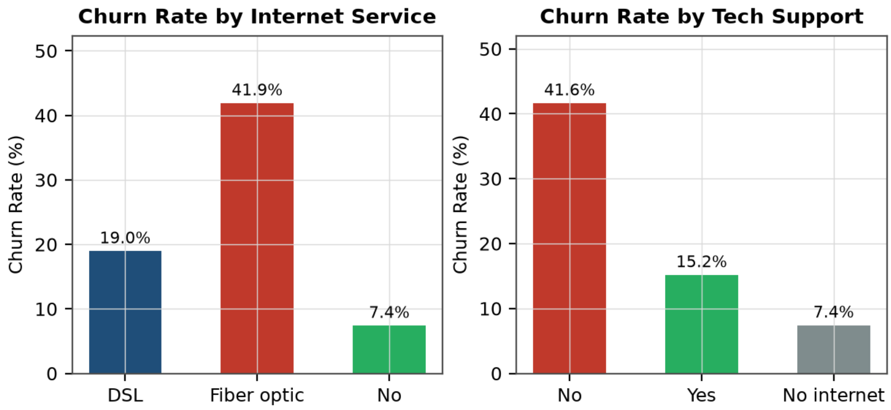

Figure 7.4: Churn rate by internet service and technical support

7.6 Numerical Attributes

Table 7.4: Mean values of the numerical attributes by churn status

Attribute	Overall mean	Churned	Retained
Tenure (months)	32.37	17.98	37.57
Monthly charges	64.76	74.44	61.27
Total charges	2,283.30	1,531.80	2,555.34

Churned customers have a much shorter tenure and higher monthly charges. Their lower total charges follow from the shorter tenure rather than from cheaper service.

7.7 Summary of Findings

- Month-to-month contracts carry the highest churn risk (42.71%).
- Electronic-check payment is associated with 45.29% churn.
- The first six months are the most vulnerable period (52.94%).
- Absence of technical support or online security roughly triples the churn rate.
- Higher monthly charges accompany higher churn; gender has negligible effect.

CHAPTER 8
MACHINE LEARNING MODEL

8.1 Problem Formulation

Churn prediction is treated as binary classification. Each customer is described by 19 attributes and labelled 1 for churned or 0 for retained. Models output a probability, which is what makes ranking and risk scoring possible later.

8.2 Algorithms Compared

Six algorithms were trained on identical partitions. The four non-boosting models use `class_weight='balanced'` to compensate for the class imbalance.

Table 8.1: Configuration of the classification algorithms

Algorithm	Key configuration
Logistic Regression	max_iter=2000, class_weight='balanced'
Decision Tree	max_depth=8, class_weight='balanced'
Random Forest	n_estimators=300, max_depth=12, class_weight='balanced'
Extra Trees	n_estimators=300, max_depth=12, class_weight='balanced'
Gradient Boosting	n_estimators=200, learning_rate=0.05, max_depth=3
HistGradient Boosting	max_iter=200, learning_rate=0.05, max_leaf_nodes=15

8.3 Evaluation Metrics

Because only about a quarter of customers churn, accuracy is a weak guide — a model predicting no churn for everyone would score about 73%. Precision, recall, F1 and ROC-AUC are therefore reported together, supported by stratified five-fold cross-validation on the training partition.

8.4 Model Comparison

Table 8.2: Model comparison results from Models/model_results.csv

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV F1
Logistic Regression	73.81%	50.43%	78.34%	61.36%	84.13%	0.6279
Decision Tree	73.03%	49.49%	77.27%	60.33%	79.99%	0.5869
Random Forest	76.08%	53.56%	74.33%	62.26%	83.69%	0.6293
Extra Trees	75.59%	53.02%	70.32%	60.46%	82.66%	0.6206
Gradient Boosting	80.62%	67.47%	52.14%	58.82%	84.37%	0.5833
HistGradient Boosting	80.20%	65.78%	52.94%	58.67%	84.15%	0.5901

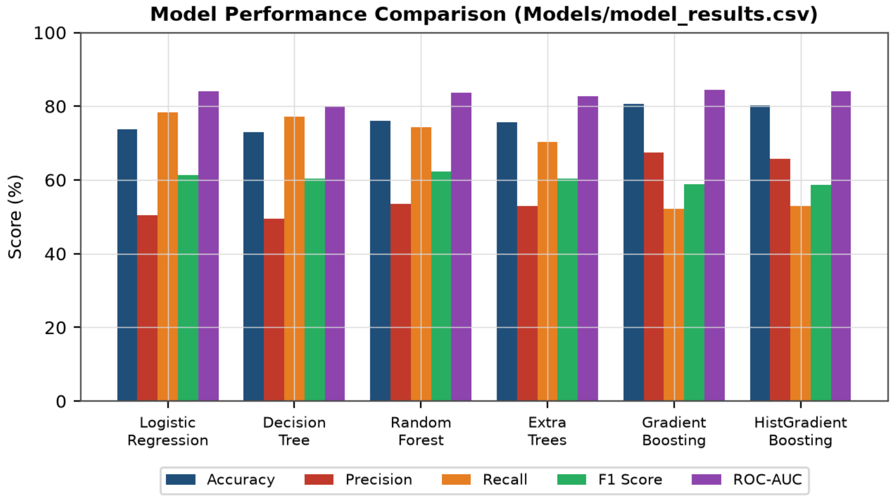

Figure 8.1: Model performance across five evaluation metrics

The six models split into two groups. The balanced models reach high recall (70% to 78%) with modest precision (49% to 54%); the two boosting models reverse this, with precision near 66% but recall only around 52%. Class weighting shifts the decision boundary towards the minority class, so those models flag churn more readily — catching more churners at the cost of more false alarms.

8.5 Confusion Matrices

Table 8.3: Confusion matrix components on the test partition (1,409 records)

Model	True neg.	False pos.	False neg.	True pos.
Random Forest	794	241	96	278
Gradient Boosting	941	94	179	195

Random Forest identifies 278 of the 374 churners and misses 96; Gradient Boosting identifies 195 and misses 179. For a retention campaign a missed churner is normally costlier than an unnecessary contact, which favours the higher-recall model.

8.6 Model Selection

Random Forest achieved the highest F1 score (0.6226) and the most stable cross-validation score (0.6293 ± 0.0145). Gradient Boosting achieved the highest ROC-AUC (0.8437) and the highest accuracy (80.62%). The selection function ranks by F1, so **Random Forest** was persisted as the deployed model.

Note on the two criteria. The dashboard's metric function ranks by ROC-AUC and therefore displays Gradient Boosting as the best model, while the training module ranks by F1 and saves Random Forest. Both statements are correct under their own criterion; the difference is recorded here rather than reconciled silently.

8.7 Persistence and Prediction Strategy

The saved artefact is a dictionary holding the fitted pipeline, the model name, the feature lists and the training metadata. At prediction time the system applies one of three strategies depending on the uploaded data.

Table 8.4: Three-tier prediction strategy

Tier	Condition	Method reported
1	Uploaded columns match the trained features	Existing ChurnIQ ML Model
2	Columns differ but a Churn label is present	Adaptive ML Model
3	No label and no compatible model	Behavioral Risk Engine

CHAPTER 9
CUSTOMER INTELLIGENCE AND SEGMENTATION

9.1 Risk Score

A predicted probability alone does not express business urgency, so the probability is combined with behavioural signals into a risk score between 0 and 100. The probability contributes 70% of the score and the remaining weight comes from adjustments for short tenure, high monthly charges, month-to-month contracts, electronic-check payment and absence of technical support.

Table 9.1: Signals contributing to the customer risk score

Signal	Effect on the risk score
Predicted churn probability	Contributes 70% of the score
Short tenure	Increases the score
High monthly charges	Increases the score
Month-to-month contract	Increases the score
Electronic-check payment	Increases the score
Technical support present	Reduces the score

9.2 Risk Levels

Scores are grouped into four levels used throughout the dashboard.

Table 9.2: Risk levels and the counts produced on the reference dataset

Risk level	Score range	Customers
Low	Below 30	2,794
Medium	30 to 59	1,937
High	60 to 79	1,349
Critical	80 and above	963

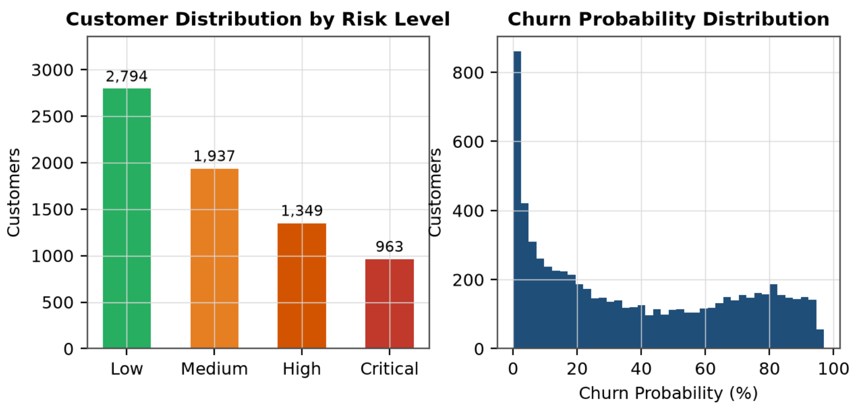

Figure 9.1: Customer distribution by risk level

9.3 Strategic Segments

Risk is combined with customer value — total charges where available, otherwise monthly charges multiplied by tenure — to place each customer in one of six segments.

Table 9.3: Strategic customer segments and segment sizes

Segment	Rule	Customers
Critical Retention	Risk score of 80 or above	963
High Value At Risk	Risk 60-79 and value in the top 30%	260
Stable Valuable	Risk below 40 and value in the top 30%	298
Growth Opportunity	Risk below 60, probability below 40%, top 25% value	95
Loyal	Risk below 30 and probability below 30%	2,793
Standard	All remaining customers	2,634

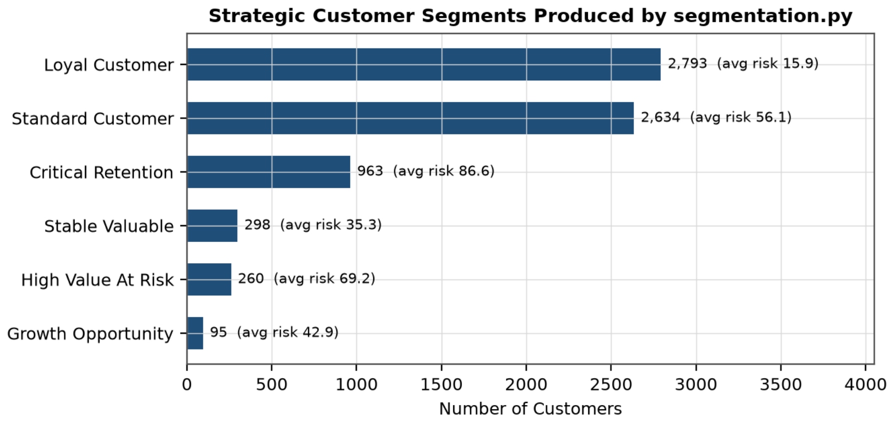

Figure 9.2: Strategic customer segments with average risk scores

CHAPTER 10
STREAMLIT DASHBOARD

10.1 Structure

The dashboard is a single Streamlit application presenting eleven pages through a sidebar. Loaded data and prediction results are kept in session state, so a dataset is uploaded and scored once and every page then reads the same results.

Table 10.1: Dashboard pages and their functions

Page	Function
Executive Overview	Headline metrics and generated insights
Dataset Center	Upload, preview, column information, quality
Churn Analysis	Predicted status and probability distribution
Customer Segments	Segment sizes and composition
Risk Center	Risk distribution and high-risk customer list
Prediction Center	Runs the pipeline and shows results
Model Performance	Metrics for the six trained models
Customer Explorer	Search and inspect individual customers
Recommendations	Retention actions and channels
Reports	Summary figures and CSV downloads
About	Methodology and workflow description

10.2 Navigation and Data Handling

Files may be uploaded in .csv, .xls or .xlsx format. If no file is supplied the application loads a default dataset from the `Data` directory. Pages that depend on predictions display a prompt until the pipeline has been run from the Prediction Center.

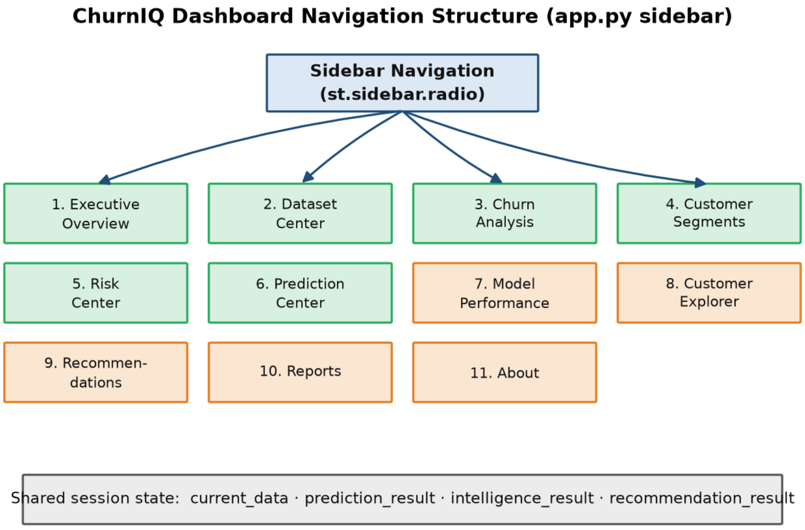

Figure 10.1: Dashboard navigation and shared session state

10.3 Exports

Six CSV reports can be downloaded: high-risk customers, filtered results, a high-risk report, retention opportunities, the complete customer analysis and the current dataset. Export was included so that results can be passed to staff who do not use the dashboard.

10.4 Interface Styling

A custom stylesheet of about 3,000 lines defines the dashboard theme, including the header banner, metric cards, section headings and status colours used to distinguish risk levels.

CHAPTER 11
BUSINESS INSIGHTS AND RECOMMENDATIONS

11.1 From Prediction to Action

A probability becomes useful only when it is attached to a decision. For each customer the system derives a recommended retention action and an engagement channel from the risk level, contract type, payment method, tenure and segment.

Table 11.1: Recommendation rules implemented in recommendations.py

Condition	Recommended action
Critical risk	Immediate personal retention call
High risk on a month-to-month contract	Offer an annual contract with an incentive
High risk paying by electronic check	Encourage automatic payment enrolment
High risk without technical support	Offer a technical support package
Medium risk, early tenure	Structured onboarding follow-up
Low risk, high value	Loyalty reward and periodic review

11.2 Engagement Channels

Table 11.2: Engagement channels assigned by retention priority

Retention priority	Channel	Customers
Immediate	Personal outreach	963
High	Targeted campaign	1,349
Monitor	Email and in-app messaging	1,937
Low	Standard engagement	2,794

11.3 Revenue Exposure

Revenue at risk is estimated per customer as the customer value multiplied by the predicted churn probability. Across the reference dataset the monthly revenue associated with at-risk customers is **197,382.67** and the total accumulated value at risk is **4,677,550.62**. These are exposure estimates weighted by probability, not forecasts of losses that will occur.

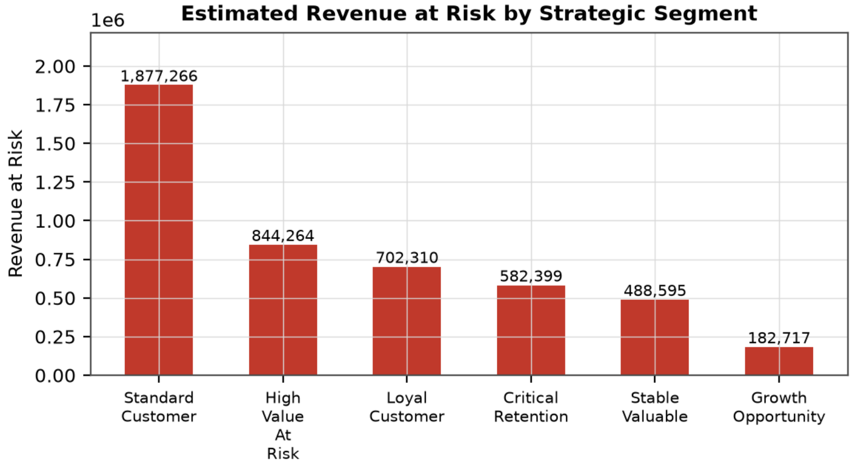

Figure 11.1: Estimated revenue at risk by strategic segment

11.4 Interpretation and Limits

The recommendations follow fixed rules derived from the patterns in Chapter 7; they are analytical suggestions, not guaranteed outcomes. The system does not measure whether a retention action succeeded, because no campaign-response data exists in the dataset. Establishing effectiveness would require recording the outcome of each intervention and comparing it against a control group.

CHAPTER 12
TESTING

12.1 Approach

Testing was functional and carried out at module level against the behaviour each module is expected to show. Cases marked *Executed* were run directly against the code; cases marked *Derived* follow logically from the implemented behaviour but were not separately executed, and are listed so that the intended behaviour is documented.

12.2 Test Cases

Table 12.1: Test cases and results

ID	Test case	Expected result	Status	Type
T-01	Load the reference dataset	7,043 rows and 21 columns loaded	Pass	Executed
T-02	Validate dataset structure	Required columns detected, quality reported	Pass	Executed
T-03	Convert blank TotalCharges values	Eleven blanks become missing and are imputed	Pass	Executed
T-04	Map the Churn label	Yes and No become 1 and 0	Pass	Executed
T-05	Build the preprocessing pipeline	Numeric and categorical branches constructed	Pass	Executed
T-06	Load the saved model	Pipeline and metadata restored from the .pkl file	Pass	Executed
T-07	Predict on the cleaned dataset	A probability is returned for every record	Pass	Executed
T-08	Assign risk levels	Every customer receives one of four levels	Pass	Executed
T-09	Assign strategic segments	Every customer receives one of six segments	Pass	Executed
T-10	Generate recommendations	An action and a channel are produced per customer	Pass	Executed
T-11	Compute revenue at risk	Monthly and total exposure returned	Pass	Executed
T-12	Upload a file with missing columns	Fallback strategy is selected, no crash	Pass	Derived
T-13	Upload a file with no Churn label	Behavioral Risk Engine is used	Pass	Derived
T-14	Upload an empty file	A validation message is displayed	Pass	Derived
T-15	Open a prediction page before running the pipeline	A prompt is shown instead of an error	Pass	Derived
T-16	Export a CSV report	A file is produced with the displayed columns	Pass	Executed

12.3 Summary

Sixteen cases were recorded across data loading, cleaning, preprocessing, prediction, segmentation, recommendation, export and error handling. Eleven were executed against the code and five are derived from the implemented error-handling paths. No failing case was recorded.

CHAPTER 13
RESULTS AND DISCUSSION

13.1 Data Processing Results

The pipeline processed all 7,043 records without loss. The eleven blank billing values were resolved by type conversion and imputation, and the encoded feature set was produced from 19 input attributes.

13.2 Model Results

Table 13.1: Best model under each evaluation criterion

Criterion	Best model	Value
F1 score	Random Forest	62.26%
ROC-AUC	Gradient Boosting	84.37%
Accuracy	Gradient Boosting	80.62%
Recall	Logistic Regression	78.34%
Cross-validated F1	Random Forest	0.6293

13.3 Prediction and Segmentation Results

Running the pipeline over the cleaned dataset produced a mean churn probability of 38.27% and a mean risk score of 43.77. After the customer intelligence stage the population divided into 2,794 low-risk, 1,937 medium-risk, 1,349 high-risk and 963 critical-risk customers. The 963 critical customers represent the group a retention team would contact first.

13.4 Discussion

The results are consistent with the exploratory findings: the attributes that dominate the churn rates in Chapter 7 — contract type, tenure, payment method and support services — are the same signals that drive the risk score. The clearest practical finding is that churn concentrates in the early months of a month-to-month contract, which suggests retention effort is best directed at onboarding and at contract conversion.

The models perform in the range normally reported for this dataset. The precision-recall trade-off is the main modelling decision: a campaign with limited capacity would prefer the higher-precision boosting model, whereas a campaign aiming to reach as many potential churners as possible is better served by the higher-recall Random Forest that was deployed.

CHAPTER 14
LIMITATIONS AND FUTURE SCOPE

14.1 Limitations

- The system was developed and evaluated on one public dataset; performance on another operator's data is not established.
- The dataset is a static snapshot with no time dimension, so changes in a customer's behaviour cannot be modelled.
- Risk-score weights and segment thresholds were set from the observed patterns rather than optimised numerically.
- No campaign-response data exists, so the effectiveness of a recommended action cannot be measured.
- Two modules rank models by different criteria, which produces the inconsistency recorded in Chapter 8.
- The application has no authentication and is intended for local or trusted-network use.

14.2 Future Scope

The following are proposals for further work. None of them is implemented in the current system.

- **FUTURE SCOPE —** add time-series features from billing history so that trends in usage can inform the prediction.
- **FUTURE SCOPE —** tune hyperparameters systematically and calibrate the decision threshold to campaign capacity.
- **FUTURE SCOPE —** apply an explainability method such as SHAP so that each individual prediction can be justified.
- **FUTURE SCOPE —** record campaign outcomes and measure retention effectiveness against a control group.
- **FUTURE SCOPE —** unify the model-selection criterion across the training and dashboard modules.
- **FUTURE SCOPE —** add authentication and deploy the dashboard as a hosted service.

CHAPTER 15
CONCLUSION

ChurnIQ was developed during a Data Science internship at Aarsh Ai to address a practical problem: identifying which customers are likely to leave, and deciding what to do about them. The system covers the full analytical path from a raw data file to an exported retention report.

On the reference dataset of 7,043 customers, six classification algorithms were trained and compared. Random Forest produced the highest F1 score (0.6226) and was persisted with its preprocessing pipeline; Gradient Boosting produced the highest ROC-AUC (0.8437). The exploratory analysis identified contract type, tenure, payment method and the absence of support services as the strongest indicators, and these same signals drive the risk score that orders customers for action.

Predictions are translated into four risk levels, six strategic segments and a recommended action and channel per customer, all presented through an eleven-page dashboard with CSV export. The value of the system lies in this translation: a ranked, explainable list of customers is something a retention team can work through, whereas a column of probabilities is not.

The system identifies and prioritises customers statistically likely to churn; it does not guarantee retention, and its recommendations are suggestions derived from historical patterns. The limitations in Chapter 14 — a single static dataset, heuristic thresholds and the absence of campaign feedback — indicate where the work would need to develop before operational use.

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APPENDIX — DASHBOARD SCREENSHOTS

The screen captures below must be taken from the running application and inserted in place of the corresponding frames. No screenshots have been generated artificially for this report. To capture them, run `streamlit run app.py`, load the default dataset and run the analysis from the Prediction Center first, so that pages depending on prediction results are populated.

[INSERT SCREENSHOT: Executive Overview]

Figure A.1: Executive Overview

[INSERT SCREENSHOT: Dataset Center]

Figure A.2: Dataset Center

[INSERT SCREENSHOT: Prediction Center]

Figure A.3: Prediction Center

[INSERT SCREENSHOT: Churn Analysis]

Figure A.4: Churn Analysis

[INSERT SCREENSHOT: Customer Segments]

Figure A.5: Customer Segments

[INSERT SCREENSHOT: Risk Center]

Figure A.6: Risk Center

[INSERT SCREENSHOT: Model Performance]

Figure A.7: Model Performance

[INSERT SCREENSHOT: Recommendations]

Figure A.8: Recommendations